Recap – dealing with LLE

Liquid-liquid separation occurs at a particular temperature when two points exist at which

$$\frac{d^2G}{dx_1^{\,2}} = 0 = \frac{d\mu_1}{dx_1} = \frac{d\mu_2}{dx_2}$$

Gibbs energy is then minimised by “unmixing” into two separate liquid phases

Binodal line defines the phase boundary

Spinodal line defines the unstable region

$$N_1 = -Dc\left(1+\frac{\partial \ln g_1}{\partial \ln x_1}\right)^{n}\frac{dx_1}{dz}$$

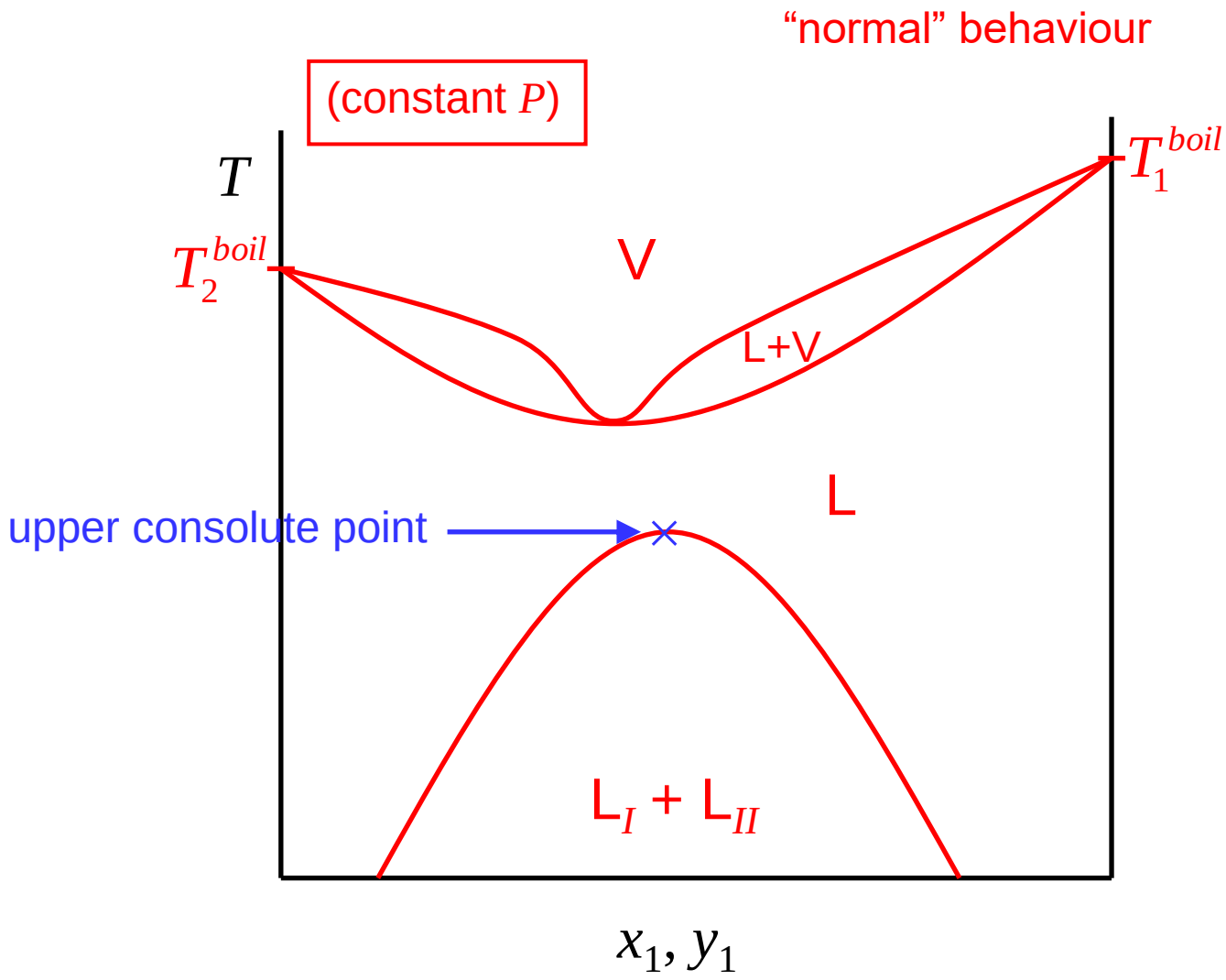
Critical opalescence

Vapour-Liquid-Liquid Equilibrium

We have seen that immiscibility can occur when the activities (and hence the fugacities) of the components in a mixture are greater than those for an ideal mixture (i.e. the activity coefficients are greater than 1)

We also know that activity coefficients greater than unity lead to positive deviations from Raoult's Law ($P > \sum x_i P_i^{SAT}$)

Liquid-liquid systems that are partially miscible therefore tend to show a maximum boiling pressure (at fixed T), or equivalently a minimum boiling temperature (at fixed P)


"normal" behaviour
(constant P)
T
T_2^{boil}
T_1^{boil}
V
L+V
L
upper consolute point
$L_I + L_{II}$
x_1, y_1

This diagram shows the situation when the bubble and dew point curves are distinct from the solubility curve

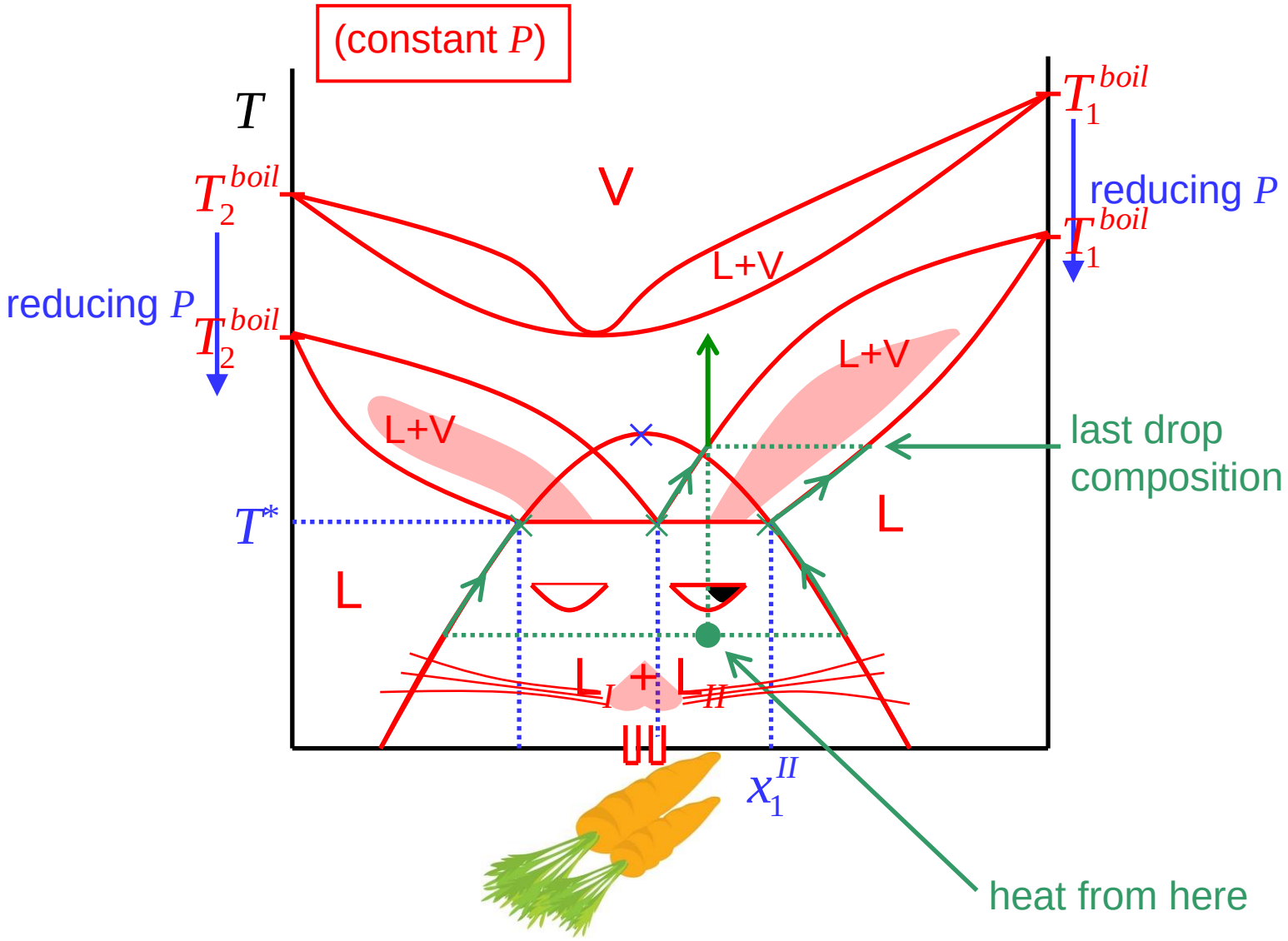
If the overall pressure is reduced, the solubility curve remains largely unaffected ($d\mu_i = \overline{V}_i dP$, and $\overline{V}_i$ is small for liquids)

whereas the bubble and dew point curves will move downwards due to the reduction in pure substance boiling temperatures

The Clausius-Clapeyron equation tells us that there is a strong dependence of T^{SAT} on P^{SAT} for pure substances at VLE

DON'T DRAW ANYTHING YET

"normal" behaviour



From the phase rule,

for a binary system with three phases present, $F=1$

So, if the pressure is fixed then the coexistence temperature for the three phases, T^*, and their compositions (x_1^I , y_1 , x_1^{II}) are also fixed

Any liquid mixture with an *overall* composition lying between x_1^I and x_1^{II} will boil (at the constant pressure) at the minimum boiling temperature T^*.

Steam distillation

$$y_1 P = \gamma_1 x_1 P_1^{SAT} \qquad y_2 P = \gamma_2 x_2 P_2^{SAT}$$

These equations must hold for **both** liquid phases (there can only be a single vapour phase)

For L_I (rich in component 2)

$$x_1^I \approx 0 \Rightarrow \gamma_2^I \approx 1 \text{ and } x_2^I \approx 1$$

so $y_2 P \approx P_2^{SAT}$

because there is only a single vapour phase

and $\gamma_1^I x_1^I P_1^{SAT} \approx P_1^{SAT} \Rightarrow \gamma_1^I \approx \frac{1}{x_1^I}$

For L_{II} (rich in component 1)

$$x_2^{II} \approx 0 \Rightarrow \gamma_1^{II} \approx 1 \text{ and } x_1^{II} \approx 1$$

so $y_1 P \approx P_1^{SAT}$

and $\gamma_2^{II} x_2^{II} P_2^{SAT} \approx P_2^{SAT} \Rightarrow \gamma_2^{II} \approx \frac{1}{x_2^{II}}$

n.b. we have assumed each liquid phase consists mainly of one component (not as drawn on phase diagram, but often true)

Overall pressure $P = y_1 P + y_2 P \approx P_1^{SAT} + P_2^{SAT}$

i.e. the vapour pressure is **elevated** in a system containing two immiscible liquids and the boiling temperature is **depressed**

Hence we can distil temperature sensitive organics at low temperature

Subsequent separation of the desired component is facilitated by liquid-liquid separation following condensation of the vapour